DEMI ZHANG

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RESEARCH INTERESTS

Bilingual sentence processing and production; memory retrieval in dependency resolution (e.g., binding) in L1 and L2; code-switching and cognitive control; computational and neural language models of human processing.

EDUCATION

University of Florida, Gainesville, FL, USA — Aug 2023 – expected May 2028
Ph.D. in Linguistics; Ph.D. Minor in Computer Science
Advisor: Dr. Edith Kaan | Minor advisor: Dr. Bonnie J. Dorr
Qualifying exam passed 2026 | GPA: 3.97/4.00

Vanderbilt University, Nashville, TN, USA — Aug 2021 – May 2023
M.Ed. in English Language Learners (Language Studies & Development track); Advisor: Dr. James R. Booth
GPA: 3.97/4.00

Nanjing Agricultural University, Nanjing, China — Sep 2016 – Jun 2020
B.A. in English; GPA: 88.86/100 (top 5%)

PUBLICATIONS

Journal Articles

Zhang, D., Pulido, E., Estrada, M. J., Kheder, S., & Kaan, E. (in press). Producing code-switches: Adaptation of cognitive control in code-switching. *Bilingualism: Language and Cognition*.

Meng, J., **Zhang, D.**, Zhong, Y., Xu, X., & Kaan, E. (2026). Comparing brain responses to moral and semantic violations. *Brain Sciences*, *16*(4), 375. https://doi.org/10.3390/brainsci16040375

Peer-Reviewed Conference and Workshop Proceedings

Singh, D., Mather, B., **Zhang, D.**, Lehman, P., Ho, J., & Dorr, B. J. (2025). FIDELITY: Fine-grained interpretable distillation for effective language insights and topic yielding. In *Findings of the Association for Computational Linguistics: NAACL 2025* (pp. 2460–2472). Association for Computational Linguistics.

Bennie, M., **Zhang, D.**, Xiao, B., Cao, J., Liu, C. X., Meng, J., & Tripp, A. (2025). PANDA – Paired Anti-hate Narratives Dataset from Asia: Using an LLM-as-a-judge to create the first Chinese counterspeech dataset. In *Proceedings of the First Workshop on Multilingual Counterspeech Generation* (pp. 1–12). Association for Computational Linguistics.

Bennie, M., Xiao, B., Liu, C. X., **Zhang, D.**, Meng, J., & Tripp, A. (2025). CODEOFCONDUCT at Multilingual Counterspeech Generation: A context-aware model for robust counterspeech generation in low-resource languages. In *Proceedings of the First Workshop on Multilingual Counterspeech Generation* (pp. 37–46). Association for Computational Linguistics.

Zhang, D., Xiao, B., Gao, C., Youm, S., & Dorr, B. J. (2024). Modeling bilingual sentence processing: Evaluating RNN and Transformer architectures for cross-language structural priming. In *Proceedings of the Fourth Workshop on Multilingual Representation Learning (MRL 2024)* (pp. 127–136). Association for Computational Linguistics.

Manuscripts Under Review and in Preparation

Jayaweera, C., **Zhang, D.**, Kempa, E., & Dorr, B. J. (under review). CLEAR-NLI: An ambiguity-first corpus for perspectivist natural language inference.

Keissar, B., Yamasaki, B., Nathaniel, U., Mathur, A., Wang, J., **Zhang, D.**, Khan, H., Bitan, T., & Booth, J. R. (under review). The neural basis of morphological learning of words [Data article]. *Data in Brief*.

Keissar, B., Yamasaki, B. L., Nathaniel, U., **Zhang, D.**, Dronjic, V., Nir, B., Booth, J. R., & Bitan, T. (in preparation). Neural mechanisms of learning and consolidation of morphologically derived words: The effect of first language similarity.

CONFERENCE PRESENTATIONS

Zhang, D., & Kaan, E. (2026, June). Who did it? Referential retrieval in L1 and L2 Spanish naturalistic listening. *Multilingual Minds and Machines Meeting (MMMM 2026)*, Max Planck Institute for Psycholinguistics, Nijmegen, Netherlands. (Poster)

Zhang, D., Pulido, E., Estrada, M. J., Kheder, S., & Kaan, E. (2025, September). Producing code-switches: Adaptation of cognitive control in code-switching. *Architectures and Mechanisms for Language Processing (AMLaP 2025)*, Prague, Czechia. (Poster)

Singh, D., Mather, B., **Zhang, D.**, Lehman, P., Ho, J., & Dorr, B. J. (2025, April–May). FIDELITY: Fine-grained interpretable distillation for effective language insights and topic yielding. *NAACL 2025*, Albuquerque, NM. (Poster)

Meng, J., **Zhang, D.**, Zhong, Y., Xu, X., & Kaan, E. (2025, March). Comparing brain responses to moral and semantic violations. *38th Annual Conference on Human Sentence Processing (HSP 2025)*, University of Maryland, College Park, MD. (Poster)

Zhang, D., Pulido, E., Estrada, M. J., Kheder, S., & Kaan, E. (2025, February). Producing code-switches: Adaptation of cognitive control in code-switching. *Florida Psycholinguistics Meeting*, Tallahassee, FL. (Poster)

Zhang, D., Xiao, B., Gao, C., Youm, S., & Dorr, B. J. (2024, November). Modeling bilingual sentence processing: Evaluating RNN and Transformer architectures for cross-language structural priming. *Fourth Workshop on Multilingual Representation Learning (MRL 2024)*, co-located with EMNLP 2024, Miami, FL. (Poster)

Zhang, D., Xiao, B., Gao, C., Youm, S., & Dorr, B. J. (2024, April). Modeling bilingual sentence processing: Evaluating RNN and Transformer architectures for cross-language structural priming. *6th Florida Linguistics Yearly Meeting (FLYM 2024)*, University of Florida, Gainesville, FL. (Talk)

Zhang, D., & Meng, J. (2024, February). Personalized teaching approaches to ESL instruction. *Pedagogy, Practice, and Philosophy 2024*, University Writing Program, University of Florida, Gainesville, FL. (Talk)

Zhang, Y., Yamasaki, B. L., & Booth, J. R. (2023, March). Examining the role of nonverbal working memory in second-language learning. *TESOL 2023 Graduate Student Research Forum*, Portland, OR. (Talk)

Zhang, Y., Yamasaki, B. L., & Booth, J. R. (2022, November). Examining the role of nonverbal working memory in second-language learning. *Vanderbilt Kennedy Center Science Day*, Nashville, TN. (Poster)

INVITED TALKS AND GUEST LECTURES

Bilingualism and second language acquisition. Invited lecture, Anyang Institute of Technology, China — May 2026
Introduction to psycholinguistics. Guest lecture, Anyang Normal University, China — May 2025
Introduction to psycholinguistics. Guest talk, Nanjing Agricultural University, China — May 2024

RESEARCH EXPERIENCE

Computational Approaches to Linguistic Meaning (CALM) Lab, University of Florida | PI: Dr. Julian Grove

Graduate Research Assistant (0.5 FTE) — Animacy inferences and pronoun processing — Aug 2026 – present

- Co-developing experiments on how verb meaning shapes probabilistic expectations about whether a novel referent is animate (*The dax sneezed* vs. *The dax melted*), and how these expectations affect later processing of *he*, *she*, and *it*.
- Designing a felicity-judgment study and a follow-up Maze experiment to test whether gradient vs. polarized animacy judgments predict the distribution of reading times at the pronoun.
- Developing stimulus sets and hierarchical Bayesian models of verb-level animacy distributions and participant-level variation.

Brain, Language, and Bilingualism Lab, University of Florida | PI: Dr. Edith Kaan

Dissertation: L1–L2 dependency resolution (supported by NSF BCS-2518248) — 2026 – present

- Testing whether L1–L2 processing differences reflect less robust syntactic representations (revised Shallow Structure Hypothesis) or differences in retrieval-cue weighting (interference account), by implementing each account as a targeted intervention in a neural language model and comparing the resulting surprisal with L1 and L2 reading-time and naturalistic-listening EEG data.

- Designed and troubleshot the E-Prime paradigm for bilingual naturalistic listening during EEG, and built Qualtrics instruments for proficiency tests (LexTALE, MELICET) and language history.
- Run EEG sessions; computed language-model surprisal for the stimuli in Python, checked forced-alignment TextGrids, and wrote R scripts for initial analysis.

Code-switching production and cognitive control (supported by NSF BCS-2017251) Jul 2024 – Sep 2026

- Led the project from conception to publication: designed a cross-task adaptation paradigm interleaving read-aloud dense code-switched or unilingual Spanish sentences with Flanker trials.
- Programmed the experiment in E-Prime, added a verbal fluency task and bilingual language profile, and developed coding schemes for switching errors, self-corrections, and hesitations.
- Analyzed articulation and conflict-adaptation data with linear mixed-effects models in R; managed IRB approval.
- Mentored two undergraduate research assistants, both co-authors on the resulting article (*Bilingualism: Language and Cognition*, in press).

Moral and semantic violations in sentence comprehension (EEG) Aug 2023 – Jan 2026

- Designed experimental sentences and study materials for an ERP study comparing brain responses to moral and semantic violations in native English speakers.
- Collected EEG data and related N400 and P600 responses to language-model-derived surprisal. Co-authored the resulting article (*Brain Sciences*, 2026).

Natural Language Processing & Culture (NLP&C) Lab, University of Florida | PI: Dr. Bonnie J. Dorr

Research Collaborator — Linguistic ambiguity in natural language inference (CLEAR-NLI) Jan 2025 – May 2026

- Designed a linguistically grounded ambiguity taxonomy and refined its categories based on pilot annotation results and inter-annotator agreement.
- Developed and revised annotation guidelines to improve consistency and reliability.
- Led the human annotation effort: recruited, trained, and coordinated annotators. Co-authored the paper introducing the corpus (under review).

Research Assistant (0.75 FTE) — Multilingual topic modeling and summarization (FIDELITY) May – Aug 2024

- Led the human evaluation of system-generated topic representations as sole annotator, assessing semantic relevance, coverage, and interpretability.
- Evaluated English topic descriptions against Tagalog word triples to assess multilingual topic generation.
- Co-authored FIDELITY (*Findings of NAACL 2025*) and presented the work at NAACL 2025 on behalf of the team.

Cross-language structural priming in neural language models Jan – Nov 2024

- Led a first-author study evaluating whether RNN and Transformer models reproduce cross-language structural priming effects observed in bilinguals (MRL 2024).

Brain Development Lab, Vanderbilt University | PI: Dr. James R. Booth

Morphological language transfer in artificial language learning (fMRI) Dec 2021 – May 2023

- Led participant recruitment and behavioral and fMRI data collection for a study of how first-language morphology shapes learning of derivational morphemes in an artificial language.
- Analyzed accuracy, reaction time, and individual differences in learning outcomes. Co-author on a data article (under review) and an empirical paper (in preparation).

Neuro-cognition of literacy in children who are deaf and hard of hearing Jun 2022 – May 2023

- Performed recruitment and data management for a study of the neural mechanisms supporting reading in DHH children.

HONORS, AWARDS, AND FUNDING

- Grinter Award, College of Liberal Arts and Sciences, University of Florida ($2,500) Feb 2026

- Conference travel support, NSF grant BCS-2518248 ($500) 2026
- Graduate Student Council International Travel Grant, University of Florida ($500) 2025
- CLAS Graduate Travel Award, University of Florida ($450) 2025
- Department of Linguistics Travel Grant, University of Florida ($450 annually) 2023 – 2026
- Outstanding Graduate Award (top 5%), Nanjing Agricultural University Jun 2020
- Government Scholarship for Study Abroad, Jiangsu Provincial Department of Education (20,000 RMB) Apr 2018
  - Funded study in the Global Leadership & Problem-Solving Summer Program, College of Liberal and Professional Studies, University of Pennsylvania Jul – Aug 2018

TEACHING EXPERIENCE

University of Florida, Gainesville, FL

Instructor of Record

ENC 1102: Argument and Persuasion Summer B 2026
ENC 3246: Professional Communication for Engineers Summer A 2026
ENC 2256: Writing in the Disciplines Summer B 2025
ENC 1101: Expository and Argumentative Writing (four semesters; 200+ students) Aug 2023 – May 2025

Teaching Assistant

LIN 2008: Language and Computers (Instructor: Dr. Sarah Moeller) Spring 2026
LIN 1140: Language and Emotion (Instructor: Dr. Eleonora Rossi) Fall 2025
ENC 3254: Professional Writing in the Discipline Summer A 2024

New Channel International Education Group, Nanjing, China Jun 2020 – Aug 2021
Full-time Teacher of ACT Science, AP Calculus, AP Physics, and IELTS Reading

- Taught ACT Science at California Crosspoint Academy, Nanjing Branch.
- Taught AP Calculus and AP Physics to students in the Sino-American program at Jinling High School.
- Gave on-campus IELTS reading lectures to undergraduates at several Nanjing universities.
- Attended weekly teaching-and-research sessions and contributed to curriculum design.
- 1,200+ cumulative teaching hours; 30+ students scored 5 on AP Calculus or AP Physics, and 50+ students scored 7+ on IELTS.

PROFESSIONAL SERVICE

Ad hoc reviewer: *Journal of Neuroscience*; ACL Rolling Review; Association for the Advancement of Artificial Intelligence (AAAI) conference
Student volunteer, NAACL 2025, Albuquerque, NM 2025
Seminar Coordinator, Graduate Linguistics Society, University of Florida Aug 2024 – May 2025
Discussion leader, Psycholinguistics Reading Group, University of Florida 2024 – 2026
EEG lab demonstrations for undergraduate psycholinguistics courses, University of Florida 2023 – 2025

COMMUNITY ENGAGEMENT

Volunteer teacher, Brain Awareness Week, Gainesville schools: Meadowbrook (2024), Glen Springs (2025), Littlewood and Westwood (2026) 2024 – 2026
Volunteer, linguistics outreach, Cade Museum for Creativity and Invention, Gainesville Nov 2025
Volunteer, Girls Do Science, Gainesville Feb 2025
Volunteer proctor and school outreach, North American Computational Linguistics Open Competition (NACLO) 2024

VOLUNTEER INTERNSHIP

FindingFive (nonprofit behavioral research platform), remote | Volunteer Data Engineering Intern Mar – Aug 2025

- Wrote SQL queries and built Apache Superset dashboards to analyze participant engagement (completion, dropouts, timeouts) and inform study-design policy.

SKILLS

Languages: Mandarin Chinese (native), English (proficient), Spanish (conversational)
Experimental methods: EEG/ERP (lab and naturalistic listening), fMRI data collection; E-Prime, Qualtrics, Gorilla; IRB protocols
Statistics and programming: R (linear mixed-effects models, hierarchical Bayesian models, ggplot2), Python (pandas, NumPy), SQL
Computational: deriving language-model predictors (surprisal, entropy) for reading-time and EEG analysis; PyTorch, Hugging Face Transformers; annotation design
Certificate: NVIDIA Deep Learning Institute, Fundamentals of Deep Learning (2024)